

Before Selecting Evolving LLM Agents: Auditing Exposure, Score Validity, and Candidate Support

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Abstract

Test-time LLM agents retain natural-language state, score responses, and revise workflows while solving tasks. Before adding a selector or privacy mechanism, we need to know whether state carries sensitive content, whether scores track executable utility, and whether generation exposes useful alternatives. We introduce a persistence-aware executable audit and apply it to a released multi-agent runtime on PlanCraft. A three-arm study detects no post-input canary exposure in 20 instances, with an upper confidence bound of 13.9%. Removing a score term assigning 85% weight to output length changes the score scale across DeepSeek and Qwen, while a secondary synthesis of 42 paired runs changes exactness by only +0.024 (95% interval $[-0.095, 0.143]$). Across four frozen cohorts, selectors restricted to recorded candidates have at most 0–8.3 percentage points of headroom. Five independent Qwen trajectories raise executable-oracle accuracy from 0.60 to 0.80 at triple the token cost. A frozen same-task public-only control reverses our initial explanation on two selected DeepSeek failures: one invalid end-point remains concentrated under both repair-conditioned and public-only prompts. These results define an audit order for evolving LLM-agent selection and its limits; they do not provide end-to-end privacy or a general prevalence estimate.

1 Introduction

Language-model agents increasingly adapt while solving a task. They revise role prompts, summarize natural-language memories, assign contribution scores, and change workflows or communication structure. TacoMAS exemplifies this design: a fast loop produces and evaluates agent outputs, while a slower controller can update feedback and topology (Xu et al., 2026). Such systems invite a natural privacy proposal. If an input can enter

persistent evolved state, privatize the scores or decisions that update that state.

This proposal joins two questions that need not have the same answer. A mechanism can provide a valid event-level DP guarantee for a bounded score while protecting a decision with negligible task value. Conversely, a useful candidate selector says nothing about whether natural-language memories and messages are private. Before measuring a privacy–utility frontier, three empirical premises must hold:

1. sensitive information measurably enters post-input persistent state
2. the score being privatized causally tracks executable utility
3. the realized candidate pool contains a better output worth selecting.

Architecture diagrams do not establish any of them. A model may fail to retain a tested secret over a short horizon. A “contribution” score may mostly reward verbose text. A perfect selector cannot recover a candidate the generator never produced.

We develop a persistence-aware executable audit around these premises and apply it to a frozen TacoMAS runtime on PlanCraft, a tool-using crafting benchmark with deterministic action-list evaluation (Dagan et al., 2025). The audit is designed to distinguish a privacy-relevant state channel from score validity and candidate-support failures. In this setting, we did not detect the motivating post-input canary channel, and removing the dominant score term did not reliably improve exactness. Most failed runs instead contained no executable candidate for any selector restricted to the recorded support to recover.

The resulting audit has a narrow systems conclusion. Across four frozen cohorts, an oracle over all persisted candidates improves accuracy by at most

0–8.3 points. Repeated repair under fixed execution feedback can concentrate on the same wrong response despite stochastic decoding. Independent trajectories expand task-relevant support and raise the pool oracle by 20 points, although at substantial cost. The latter is evidence about generation support; it is not an end-to-end privacy result. For the evaluation of test-time LLM-agent pipelines, our contributions are:

- a persistence-aware executable audit that separates exposure, signal validity, and candidate support
- a realized-pool upper bound that applies to every private or non-private selector restricted to the recorded candidates
- prospective evidence that independent generation expands support, with a matched-slot analysis locating the gain in cross-context coverage
- a scoped diagnostic on two selected hard failures showing task-level response collapse under both repair-conditioned and public-only prompts.

We do not present a new end-to-end private agent or a general leakage estimate. The result is an audit of the prerequisites that determine whether selection is worth studying.

2 Auditing Selection in Test-Time LLM Agents

2.1 Persistent exposure after the input disappears

At round t , agent v has persistent text-bearing state

$$s_{v,t} = (\phi_{v,t}, m_{v,t}, f_{v,t}, o_{v,t}, q_{v,t}),$$

covering policy text, memory, feedback, stored outputs, and score history. A canary placed in the round-one query trivially occurs in that query and may be echoed immediately. We therefore define leakage operationally as exact or preregistered five-token partial recovery from persistent state at rounds 3 or 5, after the direct-input checkpoint.

The audit uses three five-round arms: evolution with a canary, a depth-matched reset that restores initial state after each round, and evolution without a canary. The reset control distinguishes persistence created by state evolution from ordinary processing of a repeated task. Structural graph

edits are disabled so that all arms have the same three-agent topology and depth. We snapshot every declared text channel, store only match metadata in research artifacts, and separately verify the extractor by deterministic channel injection and a live memory-write positive control.

2.2 Signal validity before privacy calibration

The released non-slow-round score is

$$c(y, e) = 0.85 \min\{1, |y|/800\} + 0.15 \mathbf{1}\{e = \text{success}\}.$$

Length can therefore move the score much more than environment success. To distinguish correlation from consequence, we prospectively compare the released score with a length-zero arm while holding prompts, topology, rounds, feedback opportunities, and structural masks fixed. References are withheld until generation finishes, and deterministic executable exactness is computed offline. This firewall prevents an evaluator from leaking the answer into stopping, retention, or feedback.

The runtime authors clarify that this formula is a low-cost heuristic for fast rounds without a slow update, intended as a coarse proxy for analytical effort. During a scheduled slow update, a separately configured Judge LLM produces an independent score; the Meta-LLM receives that score together with historical heuristic and judge scores when making topology, edge, feedback, and workflow decisions. Thus the heuristic enters subsequent system state rather than serving only as a log. The authors report no independent validation for the specific 0.85 weight. We therefore treat the formula as an unvalidated design choice whose causal effect on the recorded score can be measured, rather than as a validated semantic utility estimator.

For a randomized score intervention Z , define its arm-level utility consequence as

$$\widehat{\delta}_U^{\text{abl}} = \widehat{\mathbb{E}}[U \mid Z = 1] - \widehat{\mathbb{E}}[U \mid Z = 0].$$

This randomized-arm diagnostic asks whether the intervention changes downstream utility. It is not a parameter of a candidate selector. For two candidates with utility-score gap $g_U^{\text{sel}} = u(c^+) - u(c^-)$ and candidate-score sensitivity Δ_u , define the separate local selector ratio $\rho_U^{\text{sel}} = g_U^{\text{sel}}/\Delta_u$. Private discrimination depends on $\varepsilon \rho_U^{\text{sel}}$. For a monotone exponential mechanism over k candidates, the preferred-candidate probability is

$$\frac{1}{1 + (k-1)e^{-\varepsilon \rho_U^{\text{sel}}}}.$$

If the intervention changes scores but not utility, increasing ε cannot make the targeted signal useful.

2.3 Candidate-pool headroom

Let f_i be run i 's final answer and let $\mathcal{C}_i$ contain f_i plus all persisted round-agent candidates available to a downstream selector. For executable utility U , define

$$H_i = \max_{c \in \mathcal{C}_i} U(c) - U(f_i).$$

H_i is an exact upper bound for the realized run: no selector whose output must belong to $\mathcal{C}_i$ can improve by more than H_i . This includes noisy maximum, the exponential mechanism, and any non-private reranker over the same support. The bound does not constrain regeneration, repair, or new tool interaction because those operations change $\mathcal{C}_i$.

We parse candidate text into canonical crafting plans and execute it without reference answers. The reference sequence is retained only for offline secondary checks. This distinction matters because a plausible explanation is not a successful plan, while an executable candidate with different surface text is useful.

2.4 Inference and experimental sequence

Paired binary effects use task-level risk differences, paired task bootstraps, and exact discordant-pair tests when positive and negative differences are both possible. Nested pools satisfy $\mathcal{C}_1 \subset \mathcal{C}_5$, so their support and oracle differences cannot be negative. We report task-bootstrap intervals and one-sided lower bounds, not sign-test p -values. Provider calls, rounds, candidate slots, and repeated repairs are never treated as independent task units.

The experiments proceed in a fixed order. We first audit canary exposure and cached score validity with DeepSeek V4 Flash. We then replicate the length ablation on a non-degenerate DeepSeek cohort and an untouched Qwen3.7-Max cohort. A zero-call oracle audit measures fixed-pool headroom. Finally, a prospective Qwen study expands candidate support with independent trajectories, followed by a compute-matched DeepSeek gate comparing an independent restart with direct executor-guided repair. Protocols, stopping rules, amendments, and provider-accounting details appear in the appendix.

For transparency, the appendix begins with an endpoint-status table that identifies which quantities were frozen before provider execution and

which later analyses are explicitly secondary or qualified. No post-outcome qualification changes a recorded run or its primary cohort endpoint.

3 Results

3.1 Leakage diagnostic and score validity

All 60 cells in the 20-instance canary study complete five rounds. We detect no exact or five-token partial canary in post-input persistent state in any arm (Table 1). The evolve-arm one-sided 95% upper confidence bound is 13.9%, so the study is a low-powered failed-to-detect diagnostic rather than a gate capable of excluding rare or modest exposure. The extractor recovers 7/7 deterministic channel injections and a live canary written into memory at round 2. State is active rather than inert: 617/720 comparable agent transitions change, all 168 recorded slow-feedback fields are nonempty, and retention fires in 60/300 rounds. The defensible result is therefore *failed to detect with a functioning instrument*.

The score audit also separates score movement from task movement. Scores at least 0.8 occur mostly in failed runs because failures dominate the cohort; relative to the base rate, high scores are only mildly enriched among successes. The conditional AUC analysis is reported in Appendix D rather than used as a primary result because only five runs are exact and they come from three task instances.

Randomized intervention supplies the causal result. Removing length sharply reduces mean contribution score in every study: from 0.429 to 0.069 in the initial DeepSeek pairs, by -0.308 $[-0.381, -0.230]$ in the non-degenerate DeepSeek replication, and by -0.117 $[-0.150, -0.084]$ in Qwen. It also eliminates every score at least 0.8 in the paired cohorts. Exactness does not move consistently (Table 1). The pooled direction is near zero, and the two healthy-regime point estimates have opposite signs. Length controls the released score scale, but deleting it is not a validated utility improvement.

3.2 Fixed-pool selection has little realized headroom

We execute the final answer and every persisted round-agent candidate in four independently frozen original or control cohorts. Table 2 reports the final accuracy and candidate-pool oracle. In every cohort, average headroom is between zero and 8.3 points. This is not a confidence statement about all

Table 1: Primary audit diagnostics. Exposure is measured only after the input round. Length ablations report length-zero minus released-score exactness. The 42-pair synthesis is a secondary outcome-aware analysis, not a single preregistered trial.

Study	n	Primary observation	95% interval
Canary: evolve	20	exact/partial 0/20	upper bound 0.139
Canary: reset	20	exact/partial 0/20	upper bound 0.139
DeepSeek initial ablation	10 pairs	exactness 0.000	[0, 0]
DeepSeek healthy cohort	12 pairs	exactness +0.167	[0, 0.417]
Qwen healthy cohort	20 pairs	exactness -0.050	[-0.250, 0.150]
Stratified synthesis	42 pairs	exactness +0.024	[-0.095, 0.143]

future tasks. It is an exact statement about what any selector could have recovered from each recorded pool.

Pooling the four heterogeneous original/control cohorts gives 3 recoverable failures among 65 semantically eligible runs. The two-sided Clopper–Pearson interval is [0.010, 0.129] only under an exchangeable-Bernoulli reading. The cohort-stratified bootstrap interval is [0, 0.108]. These population intervals do not alter the exact realized-pool bound.

The Qwen audit makes the failure mode concrete. Of nine failed final answers under released scoring, only one run contains an executable intermediate candidate. The other eight are generation failures. The length-zero arm similarly has one recoverable failure and nine generation failures. Most failed runs are not cases where a good candidate was scored badly. No good candidate exists on the recorded surface.

Literal duplication is model-dependent and therefore not the general explanation. Qwen repeats exact text in roughly 81–82% of its 16 slots, whereas DeepSeek duplicate-text rates are much lower. Task-relevant support is small in both model families: across selected arms, 38/72 runs contain no parseable canonical plan, 22 contain one, eight contain two, and only four contain at least three. The common bottleneck is effective executable support, not identical wording.

3.3 Independent contexts expand support

We test the bound’s escape clause prospectively on 20 untouched Qwen length-two tasks. Each task receives one temperature-zero trajectory and five independent temperature-0.7 trajectories. Every five-round trajectory supplies 15 agent outputs and a final answer. After canonical deduplication, C_j is the nested union of the temperature-zero trajectory and the first j stochastic trajectories.

Moving from C_1 to C_5 adds 1.00 canonical plan

per task, with two-sided task-bootstrap interval [0.50, 1.60] and one-sided 95% lower bound 0.60. Executable-oracle accuracy rises from 0.60 to 0.80, a gain of 0.20 [0.05, 0.40]. Because these pools are nested, we do not report sign-test p -values. A non-nested comparison between the first temperature-0.7 trajectory and the temperature-zero trajectory finds a support difference of -0.05 [-0.35, 0.25]. Temperature alone is not sufficient. The measured gain comes from accumulating independent trajectories.

At a matched 16 candidate slots, sampling across the five stochastic trajectories yields expected oracle coverage 0.626, compared with 0.570 for 16 slots within one trajectory. The difference is 0.056 [0.016, 0.105]. This cached comparison locates useful diversity across independently restarted contexts, not merely across more candidate slots within one shared context. It does not isolate a multi-agent advantage because every trajectory uses the same three-agent runtime.

The final answers alone expose the same allocation effect: an oracle over five independent stochastic finals reaches 0.65, compared with 0.570 for 16 slots sampled within one trajectory. Within this runtime, independent restarts therefore contribute more observed coverage than adding rounds and agents inside a shared trajectory, but this comparison does not identify an architectural multi-agent effect.

The gain is uneven. Across the 100 stochastic trajectories, per-task success estimates are exactly zero or one for 14/20 tasks. Four tasks are rescued by C_5 but not C_1 , and three of these contain a valid plan in only one of five trajectories. A fresh-draw plug-in expectation is 0.751 at five stochastic trajectories, whereas the realized nested C_5 pool reaches 0.80 on the same sample. The former averages estimated future draws, while the latter is the observed cached pool. The plug-in curve rises to 0.798 at twenty and has an empirical asymptote of 0.80, but

Table 2: Executable headroom for selectors restricted to persisted candidates. Intervals resample tasks within cohort. DeepSeek X1 has $n = 13$ because forward execution cannot certify the seven IMPOSSIBLE references. Diversity summaries retain all 20 evolve runs.

Cohort	n	Final	Oracle	$\hat{H}$ [95% interval]
DeepSeek X1 evolve	13	.154	.154	.000 [0, 0]
DeepSeek healthy evolve	12	.500	.583	.083 [0, .250]
DeepSeek retention control	20	.000	.050	.050 [0, .150]
Qwen released score	20	.550	.600	.050 [0, .150]

this is finite-sample extrapolation. Zero successes in five trials still permits a one-sided 95% binomial upper bound of 0.451, so 0.80 is not an identified hard ceiling.

The escape is also expensive. C_1 records 4.81M tokens and C_5 14.43M, a 3.00-fold increase for a 0.20 oracle gain. Independent restarting is an existence proof and a cost baseline, not an efficient method.

3.4 The obvious cheap remedy does not diversify hard failures

We next ask whether direct repair can beat the restart baseline without paying for another full trajectory. Eight untouched DeepSeek tasks first receive one shared temperature-zero trajectory. If its pool succeeds, both methods stop. Otherwise, one arm runs a full temperature-0.7 restart, whose realized token use caps sequential temperature-0.7 repairs in the other arm. Repair receives the public task, its current candidate, and reference-free execution diagnostics.

Five of eight anchors already succeed, leaving only three informative failures rather than the pre-registered minimum of four. The gate stops and no main cohort runs. Among those three failures, restart succeeds on 0/3 and repair on 1/3, with one discordant task and exact paired $p = 1$. On the rescued task, repair succeeds on its first 1,008-token call while the matched-budget restart fails after 157,910 tokens, an existence result on one task rather than an efficiency ratio.

The two exhausted repairs tell a different story. On TEST0030, 69/80 calls return the identical invalid string ["IMPOSSIBLE"]. After semantic normalization, 78/80 express impossibility. TEST0535 returns ["redstone_torch"] in 76/80 calls, repeatedly omitting the required intermediate. Thus nominally stochastic decoding under a fixed task, candidate, and diagnostic context concentrates on one wrong answer at raw modal rates of 86.25% and 95%.

We froze a same-task control before making another provider call. DeepSeek then received only the public task and the same output-format instruction, with no current candidate, executor diagnostic, attempt index, or response history. TEST0030 returned ["bookshelf"] in 79/80 calls (98.75%). TEST0535 returned raw ["redstone_torch"] in 76/80 calls and the same semantic plan in 80/80. None of the 160 public-task-only responses was executable. The mean task-level conditioned-minus-public-only raw modal-rate difference is -0.0625 .

This control reverses our initial explanation. Conditioning is not necessary for the concentration and does not increase it on either selected task. We therefore treat this as a scoped task-level response-collapse diagnostic: direct stochastic generation remains trapped on one invalid endpoint under both prompts. The repeated calls estimate two within-task response distributions, not 160 task units, so the result does not estimate a population prevalence. It provides a mechanism example for why raising temperature inside a direct prompt can add little support while full independent trajectories sometimes escape.

3.5 A held-out planning component control

To separate model choice from the executable support supplied by the planner, we ran a frozen 18-task diagnostic that removed bounded pre-checks and distance filtering while retaining local action verification. The unfiltered DeepSeek arm produced executable replays on 8/12 feasible tasks and canonical exact plans on 5/12; a zero-call symbolic search on the same cohort produced 12/12 and 8/12. The four replay discordances all favored symbolic search (two-sided exact $p = 0.125$); canonical exactness was 5/12 versus 8/12 ($p = 0.375$). This fixed-cohort component result localizes much of the coverage to symbolic constraints, rather than establishing an LLM or multi-agent architectural advantage.

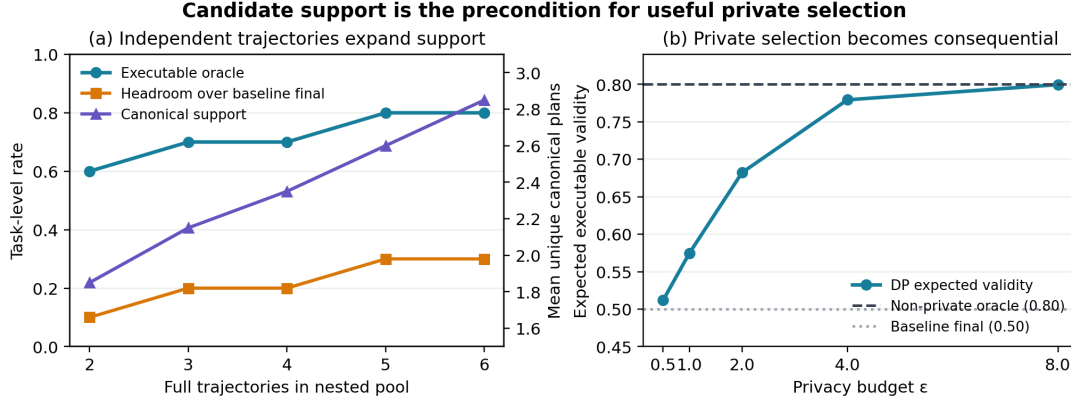


Figure 1: Independent trajectories enlarge canonical support and executable headroom. The selector calculation in panel (b) is an illustrative narrow-adjacency calibration over cached candidates, not an end-to-end private-system accuracy result. The dashed 0.80 line is the cached-pool oracle.

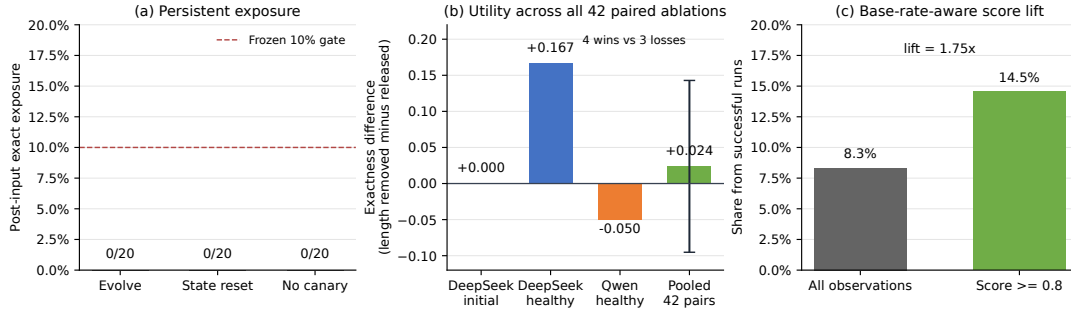


Figure 2: Secondary audit results. Post-input exposure is not detected. The 42-pair utility synthesis combines all three length interventions rather than only the initial cohort, and the original score has modest base-rate-aware lift.

3.6 Support expansion determines selector headroom

On a mixed C_5 pool, trusted binary executor utility has $g_U^{\text{sel}} = \Delta_u = 1$ and hence local $\rho_U^{\text{sel}} = 1$. Under candidate-score replacement adjacency, the monotone exponential mechanism samples index j with probability proportional to $\exp(\varepsilon u_j / \Delta_u)$. Its mean executable probability is 0.512, 0.575, 0.682, 0.779, and 0.800 at $\varepsilon = 0.5, 1, 2, 4, 8$, respectively, against the non-private oracle of 0.80. At $\varepsilon = 0.5$, 0.512 is barely above the 0.50 baseline. The curve becomes materially separated near $\varepsilon = 2$ in this binary-gap setting.

This is an illustrative selector calculation, not an experiment establishing a privacy-utility frontier. Candidate texts, prompts, memories, inter-agent messages, and control flow remain trusted; only the released selected index or output is covered by the stated adjacency. The calculation shows how a bounded selector would behave after a mixed-validity pool exists. It does not establish end-to-end privacy or demonstrate that the mechanism improves the evolving agent.

4 Related Work and Implications

Test-time optimization uses language models as optimizers or textual-gradient generators (Yang et al., 2023; Yuksekgonul et al., 2024). Self-consistency, tree search, and compute-aware inference instead spend additional test-time computation on diverse samples or structured exploration (Wang et al., 2023; Yao et al., 2023a; Zhou et al., 2023; Snell et al., 2024). Our headroom measurement complements this literature by asking a prior question: before optimizing search or selection, did the recorded generator expose any executable alternative?

Feedback-based agents such as ReAct, Reflexion, Self-Refine, and LATS condition later generation on actions, evaluation, or verbal memory (Yao et al., 2023b; Shinn et al., 2023; Madaan et al., 2023; Zhou et al., 2023). Multi-agent sampling and layered aggregation seek gains from multiple response sources (Li et al., 2024; Wang et al., 2024; Xu et al., 2026). Our matched-slot result distinguishes independent-context coverage from addi-

tional slots within one trajectory. The repair gate and public-only control show the same wrong mode under feedback-conditioned and direct prompts. Context independence is therefore an empirical variable, not an automatic consequence of temperature.

Learned evaluation can track preferences while exhibiting position, self-enhancement, and verbosity biases (Zheng et al., 2023). Length-controlled evaluation directly targets the latter confound (Dubois et al., 2024). We instead intervene on a score used inside an evolving runtime and measure machine-executable utility. The separation between score movement and task movement is therefore causal and operational, not only a meta-evaluation correlation.

Canary methods test whether rare secrets influence model outputs (Carlini et al., 2019). Our adaptation targets retained test-time state rather than trained parameters and separates immediate echo from later exposure. Work on indirect prompt injection, memory poisoning, tool-using-agent attacks, and privacy-norm behavior demonstrates that agent state and external observations can carry security- or privacy-relevant information (Greshake et al., 2023; Chen et al., 2024; Debenedetti et al., 2024; Shao et al., 2024). These threat models motivate measurement but do not imply that our specific canary must persist.

Differential privacy calibrates randomized releases to sensitivity (Dwork et al., 2006; Dwork and Roth, 2014). Report Noisy Max (Dwork and Roth, 2014), the exponential mechanism (McSherry and Talwar, 2007), and selection from candidates whose score functions are themselves private (Liu and Talwar, 2019) cover distinct private-selection settings. Private prompt optimization (Hong et al., 2024) and continual-observation tools (Dwork et al., 2010) protect other bounded channels, but formal privacy does not establish that the protected decision improves task outcomes.

The observed support bottleneck also changes the engineering target. Better reranking cannot solve generator-unreachable tasks. Independent restarts do, but at three times the token cost. Direct repair can exploit execution feedback and stop early, yet the stopped gate shows that stochastic generation can remain trapped on one invalid endpoint with or without repair conditioning. Refinement methods therefore cannot assume that feedback supplies diversity. Efficient diversity requires an intervention that changes task-relevant support

without multiplying full trajectories. Designing and validating such an intervention is future work, not a post-hoc addition to this study.

An end-to-end private architecture would still need an explicit privacy principal, adjacency relation, public transcript, and trusted boundary. Natural-language memory, cross-agent messages, stopping time, and topology changes can reveal information independently of a private score. One possible direction is a typed-edge billboard design in which cross-boundary communication is mediated through a privatized summary and slow-loop actions are post-processing. The billboard lemma gives a standard route from a shared DP signal plus each participant’s own data to joint DP outputs (Hsu et al., 2014). We do not evaluate that architecture here.

5 Reproducibility

Every reported experiment uses a frozen JSON protocol, a resumable runner, and machine-readable outputs. References are withheld during online generation and exactness is recomputed offline. Repeated calls, trajectories, and candidate slots are clustered by task in uncertainty analyses. Raw canaries and provider credentials are excluded from research artifacts. The appendix records amendments, stopping decisions, and excluded provider-error traces.

6 Conclusion

Selection in evolving LLM agents should be preceded by an audit of exposure, score validity, and candidate support. In our setting, the canary study is a low-powered failed-to-detect diagnostic, removing the dominant score term does not reliably change utility, and fixed candidate pools leave at most 0–8.3 points of realized headroom. Independent trajectories raise the oracle from 0.60 to 0.80 but triple token use. A two-task diagnostic shows task-level response collapse under both repair-conditioned and public-only prompts. These measurements define the empirical conditions that a future private selector must satisfy and expose the limits of the present evidence.

7 Limitations

The leakage audit covers one canary family, injection route, model, five-round horizon, and fixed topology. Topology evolution is disabled, so the

study does not test leakage through structural edits. With 0/20 observations it cannot exclude rates below 13.9% at one-sided 95% confidence, and it does not mount adaptive extraction attacks. Qwen replicates the score intervention but not leakage.

PlanCraft exactness is deliberately strict and may ignore useful partial reasoning. The headroom result applies only to recorded candidates and binary target-reaching utility. Hidden reasoning, new repairs, and unrecorded tool interactions lie outside the bound. The four cohort estimates are modest in size, and their intervals do not justify a universal numerical ceiling.

The contribution formula is a coarse heuristic rather than an independently validated measure of analytical effort. The runtime authors do not provide an independent justification for the 0.85 length weight. Our intervention establishes its effect on the recorded score and does not evaluate whether the separate slow-update Judge LLM improves downstream decisions.

The Qwen enlargement study has 20 tasks and one benchmark. Its nested design identifies added-pool gains but not a sign-symmetric treatment test or a same-context, token-matched continuation control. Its estimated trajectory curve assumes conditional i.i.d. restarts beyond the five observed. Cost totals are provider-recorded token counts rather than wall-clock or monetary comparisons. The matched-slot result identifies cross-trajectory coverage in the cached multi-agent runtime, not a causal multi-agent advantage.

Task-level response collapse is observed on two selected exhausted DeepSeek tasks. The public-task-only control removes candidate and diagnostic conditioning, but repeated calls still characterize only two tasks and do not create 160 independent task units. The stopped repair gate has three informative comparisons and one discordance, and the successful branch stops after one draw. Prevalence and association with success therefore remain unidentified.

Finally, the study does not provide end-to-end DP. The selector experiment protects only a candidate-validity release under a narrow replacement adjacency. It leaves candidate text and the evolving transcript trusted.

8 Ethical Considerations

The experiments use a released agent runtime, a public synthetic planning benchmark, and synthetic

canaries; they involve no human participants or personal data. The audit is designed to test privacy-relevant persistence without publishing raw canary values or provider credentials. Its failed-to-detect leakage result should not be read as a privacy guarantee for the audited runtime or for LLM agents generally.

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	This appendix preserves protocol and diagnostic details omitted from the eight-page main submission. All quantities derive from the same frozen experimental records underlying the main paper.	761
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	Endpoint status and amendments	765
	The table below separates outcomes fixed before provider execution from later analytical qualifications. The qualifications preserve the frozen runs and report rather than replace the original records.	766
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Component	Frozen endpoint	Final status or qualification
Canary audit	Post-input exact or five-token partial exposure in the three-arm 20-instance study	0/20 in each arm; reported as failed to detect, with the one-sided 95% upper bound and low power made explicit.
Length ablation	Paired exactness and recorded-score differences in each frozen cohort	Cohort results are primary; the 42-pair synthesis was added after cohort inspection and is labeled secondary and outcome-aware.
Fixed-pool headroom	Oracle over the recorded candidate surface	Final answers were retained in the support set by an explicit amendment; the pooled cross-cohort summary remains exploratory.
Nested trajectory pool	Support and oracle differences for $C_1 \subset C_5$	Nested sign tests were withdrawn because decreases are impossible by construction; task-bootstrap intervals and one-sided bounds were retained. The matched decomposition is slot-matched, not token-matched.
Repair gate	Restart versus repair after shared-anchor failure	The stop rule was reached with three informative failures, so no main cohort was run; the result is a stopped diagnostic.
Task-level response collapse	Same-task public-only control on the two exhausted tasks	The control was frozen before those calls; the two tasks remain selected mechanism cases, not a population sample.
Selector calculation	Closed-form exponential-mechanism probabilities on candidates	This is an illustrative narrow-adjacency calculation, not an empirical end-to-end DP experiment.

A Reference-firewall calibration failure

The initial reference-firewall implementation returned online quality zero for all expensive evaluations. The runtime stores a historical best candidate only when quality strictly improves over an initial value of zero. Constant-zero quality could therefore prevent candidate persistence independently of arm assignment. The issue was detected after the first pair and before interpreting its outcomes.

We froze an amendment that marked both cells `invalid_harness_calibration`, excluded them from every endpoint and the 3M valid-token cap, retained them in the artifact, and replaced constant zero with the same reference-free text heuristic in both arms. The complete pair was then restarted. The invalid cells used 282,399 tokens; the 20 valid cells used 2,716,894 tokens and 1,083 calls; total observed use was 2,999,293 tokens. This near-miss demonstrates that reference isolation must cover control flow, not merely omit a reference-derived metric from the final table.

B Detailed null sensitivity

With zero exposure events in 20 evolve instances, the exact one-sided 95% upper bound is 13.9%. A true event rate of 7.7% already gives an 80% chance of at least one observed event. In the paired leakage comparison, six pure evolve-only discordances

are required for two-sided exact $p < 0.05$, corresponding to a 30-point observed risk difference at $n = 20$. Under the explicit planning scenario of 5% control risk and 10% evolve risk with within-pair independence, power at 20 pairs is 1.7% one-sided and 0.45% two-sided; 80% power requires 373 and 463 pairs, respectively. These values are planning diagnostics, not estimates of the true exposure rate.

The independent execution-aware validation likewise has low sensitivity. Five pure method-only discordances would be required for one-sided exact $p < 0.05$, an observed risk difference of 0.25 at 20 pairs. Its observed interval $[0, 0.15]$ includes effects that would triple a 5% base rate.

C Development-stage execution-aware retention

The eight-pair development pilot completed all 16 five-round cells. Execution-aware retention produced one executable-valid and reference-exact final plan; the text heuristic produced none. Both arms had seven malformed final answers. The paired difference was $1/8 = 0.125$, with bootstrap interval $[0, 0.375]$, one method-only discordance, and two-sided exact McNemar $p = 1$.

Outcome	Text heuristic	Execution-aware	Difference
Executable-valid final answer	0/8	1/8	+1/8
Reference-trace exact	0/8	1/8	+1/8
Malformed final answer	7/8	7/8	0
Provider calls	449	446	-3
Recorded tokens	1,058,730	1,018,841	-39,889

On TEST0080, execution-aware scoring recognized an executable eight-action ladder plan at round 3, assigned it score 1, and retained the same candidate hash. The control assigned its candidate 0.02975 and ended with a sequence beginning with bamboo, which is not a crafted output and fails at step zero. This demonstrated the intended mechanism in one pair and justified independent validation; it did not establish utility improvement.

D Bootstrap handling

The original X1 run-level AUC is 0.755. In the 100,000-draw instance-cluster bootstrap, 3,903 draws (3.903%) contain no positive or no negative run and therefore have undefined AUC. The reported $[0.651, 0.853]$ interval is calculated over the 96,097 defined draws and must be read as conditional on resampling both classes. No numerical AUC is imputed for degenerate draws.

For the non-degenerate main cohort, each original-evolve run belongs to a distinct instance

cluster. Of 100,000 cluster draws, 51 (0.051%) are degenerate; the conditional interval over 99,949 defined draws is [0.625, 1.000].

E Provider repeatability detail

Across 20 cache-bypassed temperature-zero calls, two raw response hashes appear. The modal byte-identical rate is 65%, and pairwise raw-response disagreement is 47.9%. After parsing and action canonicalization, all 20 outputs are identical and exact. This result is prompt-specific; semantic identity is not implied for other prompts.

F Protocol and artifact index

- clustered score correction and null-power analysis: artifacts 063 and 066;
- execution-aware development and validation: artifacts 058 and 061;
- non-degenerate four-arm replication: artifacts 068 and 069;
- arm-level/selector-level notation correction: artifact 100;
- Qwen cross-model score replication: artifacts 074 and 075.
- Qwen candidate-pool enlargement: protocol 083 and artifacts 084–085.
- corrected nested-pool inference and trajectory decomposition: protocol 087 and artifact 088.
- public-task-only DeepSeek response-collapse control: protocol 096 and artifacts 097–098.

G Implementation clarification

After the source and intervention audit, the runtime authors clarified the intended role of the contribution signal. The length-based formula is used during regular fast rounds when no slow update is scheduled, as a low-cost and coarse proxy for analytical effort. During a slow update, a separately configured Judge LLM produces an independent score; the Meta-LLM receives that score together with historical heuristic and judge scores and uses them with other state information for topology birth/death, edge edits, feedback, and next-round guidance. The heuristic therefore enters subsequent state and decisions rather than serving only as a log. The authors did not provide an independent validation for the specific 0.85 length weight. This

clarification changes the interpretation of the intervention: it confirms consequential use of the released score while leaving its semantic validity an empirical question.

H Initial length intervention

The initial prospective intervention completed ten five-round pairs. Both arms solved TEST0095 and failed the other nine instances. Removing the length term changed mean contribution score from 0.429 to 0.069 and eliminated all 33 score observations at least 0.8, while exactness remained 1/10 in both arms.

Outcome	Original score	Length removed	Paired difference
Final executable exact	1/10	1/10	0.0
Correctness-regression instances	0	0	0
Mean contribution score	0.429	0.069	-0.360
Contribution scores at least 0.8	33	0	-33
Recorded input + output tokens	1,334,811	1,382,083	+47,272

The score reduction occurs in all ten pairs. The paired mean difference is -0.360, bootstrap interval [-0.500, -0.225], and exact two-sided sign-test $p = 0.00195$. These statistics validate manipulation of the score scale, not utility improvement.

I Independent execution-aware validation table

Outcome	Text heuristic	Execution-aware	Difference
Executable-valid final answer	0/20	1/20	+1/20
Reference-trace exact	0/20	1/20	+1/20
Malformed final answer	14/20	14/20	0
Provider calls	1,095	1,111	+16
Recorded tokens	2,669,935	2,604,731	-65,204

J Qwen cross-model replication detail

The dated qwen3.7-max-2026-06-08 snapshot completes the frozen eight-cell gate and all 40 paired main cells. The initial length-two gate scores 5/8 exact and is selected without fallback-tier search. In the 20-pair main cohort, original and length-zero scoring achieve 11/20 and 10/20 exact, respectively. The paired difference is -0.05 with bootstrap interval [-0.25, 0.15]; two length-zero-only and three original-only successes give one-sided exact $p = 0.8125$.

The causal score endpoint is decisive: the paired mean difference is -0.1167, interval [-0.1503, -0.0837], with lower scores in all 20 pairs. Counts at least 0.8 change from 8/300 to 0/300, paired interval [-0.80, -0.05]. Thus the frozen score-scale rule passes, while the utility rule does not. This is a cross-model replication of the measurement diag-

nosis, not evidence that length removal improves planning.

The provider’s reserved function-name policy requires renaming the read-only PlanCraft search tool to `recipe_search`; its arguments and environment semantics are unchanged in all cells. The runtime tracker reports 4,416,267 tokens and 2,292 calls over 41/48 cells. Seven completed cells have empty usage objects despite full trajectories, so the totals are lower bounds. Credentials are absent from the protocol, artifacts, and raw-output tree.

K Qwen intermediate-candidate oracle audit

Protocol 077 was frozen before candidate-content inspection and makes zero model calls. It evaluates every persisted round-agent answer (three agents by five rounds) and the synthesized final answer from each of the 40 Qwen main cells. Candidate texts are deduplicated by within-run SHA-256; only hashes, locations, lengths, and deterministic diagnostics are retained in artifact 078.

Arm	Final executable	Round-candidate oracle	Recoverable failed final	Generation failure	Duplicate slots
Original score	11/20	12/20	1/20	8/20	259/320
Length-zero score	10/20	11/20	1/20	9/20	263/320

Both semantic oracle gaps are 0.05 with paired-bootstrap intervals $[0, 0.15]$. Reference-exact gaps are identical. No finalization failure contains a valid round-5 candidate. The original-score recoverable count is below the frozen 4/20 advancement threshold, so the execution-gated selector route stops without API expenditure. This diagnostic does not prove that generation is the only bottleneck; it shows that selection among the persisted answer candidates has insufficient observed headroom for the planned method.

L Cross-cohort headroom and diversity

Protocol 080 freezes a zero-call extension to the 60-run DeepSeek X1 cohort, the 48-run non-degenerate main matrix, and the 40-run execution-aware confirmatory cohort. Semantic headroom excludes IMPOSSIBLE references because forward execution cannot certify non-existence. Diversity and reference-exact summaries retain all runs. The original frozen estimand compared the 15-round-candidate oracle directly with the synthesized final. This comparison is not an upper bound because a synthesizer can in principle create a valid final absent from its input candidates. We retain the original protocol hash and record an outcome-aware

amendment that adds the realized final to oracle support. The corrected quantity is weakly non-negative and bounds selectors that can retain the current final. In the 65 selected semantically eligible original/control runs, the observed frequency of a valid synthesized final with no valid round-agent candidate is 0/65. No model call or run outcome changes.

Across original/control arms, empirical fixed-pool selector headroom is 0/13 for DeepSeek X1 evolve, 1/12 for DeepSeek non-degenerate original evolve, 1/20 for confirmatory text retention, and 1/20 for Qwen original scoring. An outcome-aware exploratory stratified synthesis gives $3/65=0.046$, bootstrap interval $[0, 0.108]$. A two-sided Clopper-Pearson interval is $[0.010, 0.129]$ if the heterogeneous cohorts are treated as exchangeable Bernoulli trials. This is an exact realized-pool bound for selectors restricted to the final plus 15 persisted round-agent candidates; population intervals concern similar-task generalization. Offline headroom in the confirmatory text-retention arm and the prospective execution-aware-minus-control point estimate are both 0.05 on the same 20 tasks, a descriptive convergence rather than an independent mediation test.

Exact-text repetition is provider-specific. Qwen original/length-zero duplicate rates are 0.809/0.822, whereas DeepSeek arms range approximately 0.073–0.272. Canonical parseable-plan support is consistently small: among 72 original/control runs, 38 have zero unique parseable canonical plans, 22 have one, 8 have two, and 4 have at least three. The arms average 0.45–1.05 unique canonical plans per run, and 73%–91% of parseable slots duplicate another canonical plan. We therefore distinguish lexical diversity from task-relevant candidate support.

M Stratified length synthesis and completeness diagnostic

The 10-, 12-, and 20-pair length interventions contain 42 pairs. The stratified average exact difference is +0.0238, bootstrap interval $[-0.0952, 0.1429]$, with four length-zero-only and three original-only successes and two-sided exact $p = 1$. This outcome-aware secondary synthesis meets the frozen rule for describing a precise near-zero average: its interval contains zero and its upper endpoint is below 0.15.

The length component’s association with well-

formed output does not transfer uniformly. DeepSeek original arms yield well-formed-prediction AUCs 0.938 and 1.0, but Qwen yields 0.133 with the opposite mean direction. Paired malformed changes under length removal are 0, -0.167, and +0.15. Thus length may act as a model-dependent formation proxy, but it is not a validated cross-model completeness or semantic-quality signal.

N Prospective candidate-pool enlargement

Protocol 083 was frozen before inspecting any output on 28 untouched reference-length-two Plan-Craft tasks. Each task receives one complete uncached temperature-0 trajectory and five independent complete uncached temperature-0.7 trajectories. Thinking is disabled throughout. A trajectory contributes 15 round-agent answers and one synthesized final; canonical plans are deduplicated before support and execution measurement. The nested pool C_j contains the temperature-0 trajectory and stochastic repetitions 1 through j . This makes $C_5 - C_1$ a randomized generation-budget contrast over an identical task set.

The eight-task gate passes: mean $S_5 - S_1 = 0.875$, 6/8 tasks have $S_5 \geq 2$, all 48 valid trajectories complete five rounds, and all have positive usage. A provider payment interruption had previously created nine persisted traces containing Access denied or overdue-payment; an integrity back-audit marks all nine invalid, including two with partial positive usage. After seven valid tasks, the conservative 6.5M gate-token projection blocked the eighth task because those invalid attempts had consumed budget. The recorded amendment raises only this cap to 7.5M. The original and amended hashes begin 7ccf128c5bd4 and 97c35906de54; the closure manifest stores both complete values. At amendment time, nested non-negative support gains and the first seven task summaries made both substantive gate conditions mathematically irreversible. No task, condition, endpoint, threshold, or main budget changed.

The 20-task main cohort completes all 120 trajectories without an integrity failure. Results are:

Contrast	Mean difference	Task-bootstrap 95% interval	One-sided 95% lower bound	Exact test
Canonical support, $C_5 - C_1$	+1.00	[0.50, 1.50]	0.50	withdrawn: nested
Executable oracle, $C_5 - C_1$	+0.20	[0.05, 0.40]	0.05	withdrawn: nested
Headroom over baseline final, $C_5 - C_1$	+0.20	[0.05, 0.40]	0.05	withdrawn: nested
Standalone support, temperature 0.7 minus 0	-0.05	[-0.35, 0.25]	not applicable	$p = 1.0$

The nested support/oracle/headroom curves are 1.85/0.60/0.10, 2.15/0.70/0.20, 2.35/0.70/0.20, 2.60/0.80/0.30, and 2.85/0.80/0.30 for C_1 through

C_5 . The originally reported nested sign-test values are invalid because decreases are excluded by construction and are formally withdrawn in artifact 088. The non-nested standalone temperature contrast permits either direction and retains its $p = 1$ result.

The trajectory decomposition contains 57 successful stochastic trajectories among 100. The anchor-inclusive plug-in oracle estimates are 0.660, 0.700, 0.722, 0.738, 0.751, 0.784, and 0.798 for $n = 1, 2, 3, 4, 5, 10$, and 20. The $n > 5$ values assume conditionally i.i.d. trajectories and are extrapolations. Task-cluster intervals, rather than trajectory-level independent intervals, preserve the 20-task experimental unit.

Of the four C_5 -only rescues, TEST0102, TEST0105, and TEST0361 contain a valid plan in 1/5 stochastic trajectories; TEST0526 contains one in 4/5. Five stochastic final answers alone attain 0.65 oracle coverage. At a matched count of 16 slots, cached cross-trajectory mixing has expected oracle 0.626 versus 0.570 within one stochastic trajectory, difference 0.056, task-bootstrap interval [0.016, 0.105]. All trajectories use the same three-agent runtime, so this is not a single-agent comparison.

Main-cohort generation costs are 4,807,489 recorded tokens for C_1 and 14,430,644 for C_5 . The incremental four trajectories consume 9,623,155 tokens, a total ratio of 3.0017. Across gate and main, 168 valid trajectories record 10,546 calls and 20,489,906 tokens. Including the nine retained integrity-failed attempts gives 10,590 calls and 20,573,345 recorded tokens.

For cached C_5 candidates, binary valid-versus-invalid executor utility gives $g_U^{\text{sel}} = \Delta_u = 1$ and hence local $\rho_U^{\text{sel}} = 1$ whenever a mixed-validity pool exists. This candidate-level ratio is distinct from the arm-level ablation consequence δ_U^{abl} . The earlier artifact 071 conflated these estimands and its epsilon inversion is withdrawn; artifact 100 records the zero-call correction. The monotone exponential mechanism gives mean exact valid-selection probabilities 0.5119, 0.5748, 0.6822, 0.7793, and 0.7996 at $\varepsilon = 0.5, 1, 2, 4$, and 8. These values are the expected consequence of a maximally separated binary signal on a small pool, not an independently discovered frontier. Under candidate-score replacement adjacency, one candidate's clipped validity bit may change and only the selected index/output is released; candidate text and the remaining system are trusted.

O Stopped DeepSeek compute-matched generation gate

Protocol 090 freezes a non-nested restart-versus-repair comparison before any provider call on 20 untouched reference-length-two tasks: eight adaptive gate candidates and 12 reserved main tasks. Each task receives a shared temperature-0 five-round, three-agent trajectory. An executable anchor-pool candidate causes symmetric success and zero added intervention calls. On anchor failure, a new temperature-0.7 trajectory defines realized token cap B_i ; sequential direct repair candidates count toward the primary pool only while cumulative repair tokens do not exceed B_i . Repair uses the public task and reference-free executor diagnostic, never the hidden reference. Unsuccessful repair stops at its first budget crossing or 80 calls.

All eight gate tasks complete with provider integrity. Five anchors succeed, so only TEST0181, TEST0030, and TEST0535 are informative. Their outcomes are:

Task	Restart	Repair	Restart tokens	Included repair tokens	Repair calls
TEST0181	0	1	157,910	1,008	1
TEST0030	0	0	138,559	51,330	80
TEST0535	0	0	162,193	50,313	80

The frozen gate requires at least four informative anchor failures. It therefore stops even though repair has at least one rescue, does not underperform restart in rescue count, all provider records pass integrity, and both failed repair branches reach the 80-call exhaustion rule. No main task is run. Conditional on anchor failure, repair minus restart is +0.333 with task-bootstrap interval $[0, 1]$ and exact paired $p = 1$. We do not interpret an all-eight difference because five symmetric successes are inherited from the shared anchor rather than caused by either intervention.

Repair diversity is the principal diagnostic. TEST0181 succeeds immediately with ["dark_oak_planks", "pink_bed"]; because it stops after one call, its modal concentration is not estimable. TEST0030 has only three raw response forms over 80 calls: 69 ["IMPOSSIBLE"], nine IMPOSSIBLE, and two empty responses. Its raw modal rate is 86.25%, semantic normalization combines the first two forms into an IMPOSSIBLE intent in 78/80 calls (97.5%), and semantic pairwise agreement is 95.1%. TEST0535 also has three raw forms, with ["redstone_torch"] appearing 76/80 times (95% raw and semantic modal rates; 90.4% semantic pairwise agreement); the missing intermediate redstone makes it non-executable.

Across the two exhausted tasks, the call-weighted raw modal share is $145/160=90.625\%$, and the semantic share is $154/160=96.25\%$.

The repair-only 0.80 threshold was introduced after observing the stopped gate and was not a preregistered test. Calls within a task are not independent task units: the observed 86.25%–95% raw range comes from two failures only. Failure-type association is not estimable with three repair branches, and cross-cohort modal stability is unavailable because earlier disjoint DeepSeek cohorts did not record repeated direct repairs under this prompt. The executor behaves as intended in both failures: it does not certify IMPOSSIBLE, and it rejects plans that omit required intermediate products.

We then froze protocol 096 before making a same-task public-only control. Each exhausted task received 80 additional DeepSeek temperature-0.7 calls with the same output-format instruction but no candidate, executor diagnostic, attempt index, response history, or reference. TEST0030 produced raw ["bookshelf"] in 79/80 calls (98.75%); TEST0535 produced raw ["redstone_torch"] in 76/80 and the same semantic plan in 80/80. No response was executable. Both tasks exceed the preregistered 0.80 near-determinism threshold, and the mean task-level conditioned-minus-public-only raw modal-rate contrast is -0.0625. The control therefore does not support conditioning as the cause. We use **task-level response collapse** for the repeated invalid endpoint under both direct prompts. Protocol 096 treats the two tasks as the independent units and forbids a call-level population test.

The first two direct calls on TEST0030 reached the output limit in hidden reasoning and exposed empty visible answers because the direct path omitted the full runtime’s explicit non-thinking switch. Both calls, 3,768 tokens, and malformed outcomes remain in the intention-to-repair record. Subsequent direct calls pass thinking.type=disabled; no prior output is deleted or rerun. LiteLLM response caching is disabled throughout, while DeepSeek’s automatic provider-side prefix-cache accounting is separately outside that client flag. Total gate usage is 736 calls and 1,642,567 recorded tokens: 1,081,254 anchor, 458,662 restart, and 102,651 repair tokens. TEST0181 shows only that repair succeeded on its first attempt while a matched-budget restart failed; comparing first-draw tokens with a fully spent failed trajectory as a multiplicative efficiency ratio would be mis-

leading. This stopped gate is positive mechanism evidence about two observed failures, but not a positive comparative-method claim.

P Held-out planning component control

The final planning diagnostic used a frozen 18-task cohort and no new benchmark selection. The symbolic-only control reproduces 12/12 feasible replays, 8/12 canonical exact plans, six bounded stops on impossible-reference tasks, and 14/18 evaluator-exact labels. Ten of the 12 feasible cached plans are sequence-identical to the symbolic plans; the two remaining replay-valid plans use alternative action orderings. Only 7/78 accepted requests exposed multiple options, and four step-index mismatches occurred in two replay-valid tasks. These branch-level quantities are descriptive for the fixed cohort.

The prospective unfiltered DeepSeek arm removes bounded pre-checks and distance filtering but retains local action verification. It uses 142 provider calls and 43,623 recorded tokens, reaching 8/12 feasible targets with 5/12 canonical exact plans. Against symbolic search, replay has four symbolic-only and no live-only pairs (two-sided exact $p = 0.125$); canonical matching has four symbolic-only and one live-only pair ($p = 0.375$). The six impossible-reference runs terminate without reaching a target, which does not certify impossibility. This component comparison is separate from architecture-level inference and is not a causal multi-agent result.